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<https://zprivf.github.io/applied-water-research-lab/index.html>

AREA OF INTEREST

- **MODELING OF COASTAL HAZARDS**
STORM SURGE AND COMPOUND FLOODING; PROBABILISTIC ENSEMBLE SIMULATIONS; DATA ANALYTICS (ARTIFICIAL INTELLIGENCE, MACHINE LEARNING/DEEP LEARNING - PHYSICS-INFORMED/AWARE NEURAL NETWORKS); THERMOHALINE PROCESS AND CIRCULATION UNDER EXTREME STORMS; TRANSPORT OF SEDIMENT AND CONTAMINATED FLUID (UNDER EXTREME WEATHER); EXTREME VALUE STATISTICS AND RISK ANALYSIS; PREDICTION OF POTENTIAL IMPACTS OF EXTREME WEATHER CONDITIONS
- **ENVIRONMENTAL MODELING**
NUMERICAL MODELING OF COASTAL AND OCEAN HYDRODYNAMIC AND ENVIRONMENTAL PROCESSES; RIVER MODELING; MICRO PLASTIC TRANSPORT; ESTIMATION AND EVALUATION OF THE ENVIRONMENTAL EFFECTS FROM HUMAN USE (DESALINATION EFFLUENCE, POLLUTION); WATER QUALITY; LONG-TERM SCENARIOS RESPONSE TO FUTURE CLIMATE CHANGE; DATA ASSIMILATION
- **DIGITAL TWIN FOR COASTAL AND URBAN FLOOD RESILIENCE**
DIGITAL TWIN FOR COASTAL RESILIENCE AND SMART FLOOD INFRASTRUCTURE; INTEGRATION OF REAL-TIME SENSORS, HYDRODYNAMIC SIMULATIONS, AND AI-BASED FORECASTING TO ENABLE SMART ADAPTATION STRATEGIES
- **NATURE-BASED SOLUTIONS (NBS) AND CLIMATE ADAPTATION PLANNING**
NATURE-BASED SOLUTIONS FOR CLIMATE ADAPTATION; BMI DYNAMIC VEGETATION MODELING AND EVALUATING THE EFFECTIVENESS OF GREEN INFRASTRUCTURE AND ECOLOGICAL INTERVENTIONS UNDER EXTREME HYDROMETEOROLOGICAL SCENARIOS.

EDUCATION

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|--------------------------------|--|-----------|
| • Doctor of Philosophy | Civil Engineering, <u>Texas A&M University</u> , College Station, TX. USA | Aug. 2018 |
| • Master of Engineering | Civil Engineering, <u>Seoul National University of Sci. & Tech.</u> , Seoul, South Korea | Feb. 2012 |
| • Bachelor of Science | Civil Engineering, <u>Seoul National University of Sci. & Tech.</u> , Seoul, South Korea | Feb. 2010 |

APPOINTMENTS

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|---|--|-------------------|
| • Research Assistant Professor (Principal Investigator) | Bureau of Economic Geology (BEG), <u>University of Texas</u> , Austin, TX. | May.2022-present |
| • National Academies of Science, Engineering, Medicine - National Research Council (NRC) Postdoctoral Fellow | Ocean Dynamics and Prediction Branch, <u>U.S. Naval Research Laboratory (NRL)</u> , NASA's John C. Stennis Space Center, MS. | Dec.2018-Dec.2021 |
| • Research Assistant | Zachry Department of Civil Engineering, <u>Texas A&M University</u> , College Station, TX. | Jun.2014-Aug.2018 |
| • Research Assistant | Department of Civil Engineering, <u>Seoul National University of Sci. & Tech.</u> , Seoul, South Korea. | Mar.2011-Feb.2012 |
| • Teaching Assistant | Department of Civil Engineering, <u>Seoul National University of Sci. & Tech.</u> , Seoul, South Korea. | Mar.2010-Feb.2011 |
| • Hydraulic Laboratory Assistant | Department of Civil Engineering, <u>Seoul National University of Sci. & Tech.</u> , Seoul, South Korea. | Sep.2009-Dec.2009 |

MEMBERSHIP IN PROFESSIONAL SOCIETIES

- Member, American Meteorological Society (from 2023)
- Member, American Geophysical Union (from 2016)

- Member, European Geosciences Union (from 2024)
- Member, Texas A&M Energy Research Society (from 2016)
- Member, American Society of Civil Engineers (from 2014)
- Member, Korean-American Scientists and Engineers Association (from 2012)

AWARDS AND FELLOWSHIPS

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- First Author Achievement Awards 2025 in BEG, University of Texas at Austin, Austin, TX Apr. 2026
 - BEG Grants Awards, Proposal Development Assistance Grant Program, Bureau of Economic Geology, UT Austin Fall 2025
 - First Author Achievement Awards 2024 in BEG, University of Texas at Austin, Austin, TX Apr. 2025
 - Best Both Research and Poster Awards in Climate Health Frontiers 2025 Symposium, University of Houston, Houston, TX Jan. 2025
 - BEG Grants Awards, Proposal Development Assistance Grant Program, Bureau of Economic Geology, UT Austin Fall 2024
 - BEG Grants Awards, Proposal Development Assistance Grant Program, Bureau of Economic Geology, UT Austin Spring 2024
 - BEG Grants Awards, Proposal Development Assistance Grant Program, Bureau of Economic Geology, UT Austin 2023
 - BEG Grants Awards, Proposal Development Assistance Grant Program, Bureau of Economic Geology, UT Austin 2022
 - NRC Research Associateship Award, Post-Doctoral Fellowship, U.S. Naval Research Laboratory 2018-2021
 - The American Bureau of Shipping Scholarship, Ocean Eng. Scholarship Committee, TAMU 2015-2016
 - The Bechtel Foundation Services Scholarship, Ocean Eng. Scholarship Committee, TAMU 2014-2015
 - General Scholarship supported by the University Development Fund, Dept. of Civil Engineering, SNUST Fall-2011
 - The Alumnus Scholarship, Dept. of Civil Engineering, SNUST 2010-2011
 - The Academic Excellence Scholarship, Dept. of Civil Engineering, SNUST 2010-2011
 - General Scholarship supported by the University Development Fund, Dept. of Civil Engineering, SNUST Spring-2010
 - The Scholarship of Ministry of Patriots & Veterans Affairs, Dept. of Civil Engineering, SNUST 2004-2010

DEPARTMENT AND UNIVERSITY SERVICE

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- Seminar Coordinator, Bureau of Economic Geology, University of Texas, Austin, TX.
<https://www.beg.utexas.edu/presentations/seminars> Spring-2024
 - Seminar Coordinator, Bureau of Economic Geology, University of Texas, Austin, TX. Fall-2023

SERVICE

Peer-Review Activities

Review manuscripts in a variety of journals, including the following:

- Natural Hazards
- Journal of Marine Science and Engineering
- Journal of Geophysical Research: Oceans
- Journal of Waterway, Port, Coastal, and Ocean Engineering
- Journal of Geophysical Research: Atmospheres
- Earth's Future
- Estuaries and Coasts
- Journal of Hydrology
- Water Resources Research
- Communications Sustainability
- Communications Earth & Environment
- Journal of Atmospheric and Oceanic Technology
- Ocean Systems Engineering
- Marine Geodesy
- Discover Applied Sciences

Review Research Proposals in a variety of agencies, including the following:

- National Science Foundation (NSF)

Editorial Service

- Review editor - Frontiers in Water

Conference Committee

- HydroML 2026 - Organizing Committee

MEDIA COVERAGE AND PUBLIC ENGAGEMENT

Expert interview

- KXAN IN-DEPTH Investigates / Interviewed by The Hill on AI-assisted flash-flood prediction and Google's Flood Hub tool. "Google rolls out AI tool to predict flash floods up to 24 hours ahead." The Hill, 2026. <https://www.kxan.com/weather-traffic-qas/google-rolls-out-ai-tool-to-predict-flash-floods-up-to-24-hours-ahead/>

SELECTED PUBLICATIONS

Journal Articles

- Kyne, D., Scanlon B.R., Zhang, Y., Stephens K., Porter, B., Lee, W. and Vega, A. (2026) "A Parcel-Level GIS Analysis of Development Disparities, Floodplain Exposure, and Building Growth in Texas Border Colonias.", *Sustainability*, 18, 8603. <https://doi.org/10.3390/su18168603>.
- Chowdhury, M.E., Lee, J.W and Lee, W. (2026) "Spatiotemporal quantification of relative contributions of compound flood drivers using Shapley value-based approach", *Journal of Hydrology*, 677, 135808. <https://doi.org/10.1016/j.jhydrol.2026.135808>.
- Cruz, A.U.S., Lee, J.W., and Lee, W. (2026) "Pyflood: Coastal Flood Mapping with High-Resolution Digital Elevation Model and Land Cover Data", *Environmental Modelling & Software*, 203, 107010. <https://doi.org/10.1016/j.envsoft.2026.107010>
- Lee, W., Sun, A.Y. and Scanlon, B.R. (2026) "Rapid Intensification and Relative Sea-Level Rise Amplify Compound Flooding from Hurricanes Harvey and Beryl", *Earth's Future*, 14, e2025EF007678, <https://doi.org/10.1029/2025EF007678>.
- Lee, W., Sun, A.Y. and Scanlon, B.R. (2025) "Probabilistic Storm Surge and Flood Inundation Modeling of the Texas Gulf Coast Using Super-Fast INundation of CoastS (SFINCS)", *Coastal Engineering*. <https://doi.org/10.1016/j.coastaleng.2025.104721>
- Lee, W. and Veeramony, J. (2024) "The Response of Mixed Layer Depth Due to Hurricane Katrina (2005)", *Journal of Marine Science and Engineering*, 12, no. 4: 678. <https://doi.org/10.3390/jmse12040678>
- Lee, W., Sun, A.Y., Scanlon, B.R. et al., (2023) "Hindcasting compound pluvial, fluvial and coastal flooding during Hurricane Harvey (2017) using Delft3D-FM", *Nat Hazards*. <https://doi.org/10.1007/s11069-023-06247-9>
- Sun, A.Y., Li, Z., Lee, W., Huang, Q., Scanlon, B.R., Dawson, C. (2023) "Rapid Flood Inundation Forecast Using Fourier Neural Operator", In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 3733-3739)arXiv:2307.16090. <https://doi.org/10.48550/arXiv.2307.16090>
- Lee, W., McLaughlin, P.W., and Kaihatu, J.M. (2017) "Parameterization of maximum significant wave heights in coastal regions due to hurricanes", *Journal of Waterway, Port, Coastal and Ocean Engineering*, 143(2), [https://doi.org/10.1061/\(ASCE\)WW.1943-5460.0000362](https://doi.org/10.1061/(ASCE)WW.1943-5460.0000362)
- Hong, S.H., Kim, Y.K., Lee, W. and Chung, E.S. (2012) "Rainfall Variations of Temporal Characteristics of Korea Using Rainfall Indicators", *KoreaWater Resources Association*, Korea. DOI: 10.3741/JKWRA.2012.45.4.393
- Lee, W., Hong, S.H., Kim, Y.K., and Chung, E.S. (2011) "Temporal & Spatial Variability of Precipitation and Daily Average Temperature in the South Korea", *Korean Society of Hazard Mitigation*, Korea. <https://doi.org/10.9798/KOSHAM.2011.11.4.073>
- Kim, Y.K., Hong, S.H., Lee, W., and Chung, E.S. (2011) "Variations of Weather Stations Characteristics to Recent Climate Change Using Rainfall & Average Daily Temperature Data", *Korean Society of Hazard Mitigation*, Korea. UCI : G704-001845.2011.11.4.016

Submitted / Under Review

- Souza, P.M.J., Lee, W., Bolser, D.G., Souza, C.M., and Brandl, S.J. (2026) "Acoustic competition amplifies climate-driven loss of fish spawning habitat", *Nature Climate Change* (submitted, under review).
- Lee, W., Zhang, Y., Kyne, D., Stephens K., Porter, B., Vega, A. and Scanlon B.R. (2026) "Multi-Event Flood Risk Assessment and Adaptation Prioritization in Underserved Border Communities: Colonias in the Lower Rio Grande Valley, Texas", *International Journal of Disaster Risk Reduction* (submitted, under review).
- Yang, H., Lee, J.W. and Lee, W. (2026) "Where, When, and How Do Physics-Informed Deep Learning Models Improve Hurricane-Induced Compound Flooding Predictions?" *Water Resources Research* (submitted, under review).
- Zhang, Y., Scanlon B.R., Kyne, D., Stephens K., Porter, B., Dashtian, H. and Lee, W. (2026) "Decadal Settlement Dynamics in Texas Colonias Revealed by Remote Sensing", *Environmental Research Letter* (submitted, under review).

In Progress

- Lee, W., Sun, A.Y., Scanlon, B.R and Azizzadenesheli, K. (2026) "Physics-Aware Neural Operators for Compound Flood Prediction: A Multi-Model Benchmark in the Texas Gulf Coast" (in preparation)

Conference Publications

- Development of a Physics-Informed Deep Learning Model for Hurricane-Induced Compound Flooding Prediction - Where, When, and How Do Physics-informed Deep Learning Models Improve Hurricane-Induced Compound Flooding Predictions? (HydroML - Austin, Austin, TX) May.2026
- Highlighting the Needs for Machine Learning for Hydrologic Applications (HydroML Symposium, 2026, Austin, TX) May.2026
- Acoustic Niche Partitioning Accelerates Modeled Climate-Driven Spawning Habitat Loss, (Benthic Ecology Meeting 2026, Virginia Beach, VA) Mar.2026
- Understanding Climate-Driven Compound Flood Risk: The Role of Hurricane Evolution and Relative Sea-Level Rise in Hurricanes Harvey (2017) and Beryl (2024), (AGU Fall Meeting 2025, New Orleans, LA) Dec.2025
- Advanced Flood-Inundation Mapping with Physics-Aware Neural Networks and Ensemble Simulations (Scientific Machine Learning Methods for Complex Dynamical Systems, 18th U.S. National Congress on Computational Mechanics, Chicago, IL) Jul.2025
- From Harvey to Beryl: Understanding the Evolution of Compound Flooding Under Rapidly Intensifying Hurricane (Texas Climate Conference 2025, Houston, TX) Apr.2025
- Assessing Climatic Impacts on Wetland Transformation along the Texas Coast with Google Earth Engine (Texas Climate Conference 2025, Houston, TX) Apr.2025
- Nonlinear evolution and Primary Drivers of Relative Sea-Level Rise Along the U.S. Gulf Coast (Texas Climate Conference 2025, Houston, TX) Apr.2025
- Efficient Coastal Flood-Inundation Mapping Using Machine Learning-Enhanced Static Model (NHRI Computational Symposium, UCLA, Los Angeles, CA) Feb.2025
- Enhancing Rapid Flood-Inundation Prediction and Mapping with Fourier Neural Operators: A Hybrid Approach to Compound Flooding Events (Climate Health Frontiers 2025 Symposium, University of Houston, Houston, TX) Jan.2025
- Advancing Rapid Flood-Inundation Prediction and Mapping: Fourier Neural Operator Approach to Compound Flooding Events (AGU Fall Meeting 2024, Washington, D.C) Dec.2024
- Efficient Static Flood Model for Rapid Coastal Flood-Inundation Mapping (AGU Fall Meeting 2024, Washington, D.C) Dec.2024
- Rhythms of Sea Level Rise in the Gulf of Mexico: Annual Budget Analysis and Integration of Deterministic and Stochastic Forcings (1900-2024) (AGU Fall Meeting 2024, Washington, D.C) Dec.2024
- Assessing Climatic Impacts on Wetland Transformation and Resilience along the Texas Coast using Remotely Sensed Imagery (AGU Fall Meeting 2024, Washington, D.C) Dec.2024
- Lee, W., Sun, A. Y., and Scanlon, B.R (2024) Advancing Coastal Resilience through Integrated Modeling of Compound Flooding Events, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2024, EGU24-21937, <https://doi.org/10.5194/egusphere-egu24-21937> Apr.2024
- Probabilistic Storm Surge and Flood-Inundation Modeling in Texas Gulf Coast Using Super-Fast INundation of CoastS (SFINCS) (AGU Fall Meeting 2023, San Francisco, CA) Dec.2023
- Rapid Flood Extent Forecasting Using Hybrid Flood Inundation Modeling and Deep Learning (AGU Fall Meeting 2023, San Francisco, CA) Dec.2023
- Development of a hybrid hydrodynamic and deep learning model for coastal flood inundation prediction (HydroML Symposium, 2023, Berkeley, CA) May.2023
- Modeling of Storm-induced Surge and Flooding from Hurricane Harvey in 2017 (Planet Texas 2050, Spring Research Symposium 2023, Austin, TX) Mar.2023
- Modeling of Storm-induced Surge and Flooding from Hurricane Harvey in 2017 (AGU Fall Meeting 2022, Chicago, IL) Dec.2022
- Predictions of Waves and Water Levels in Puerto Rico using a Coupled Wave-Current Modeling System (Frontiers in Hydrology Meeting 2022, San Juan, Puerto Rico) Jun.2022
- Hydrodynamic Predictions along the Puerto Rican coast during Hurricanes Humberto, Jerry, and Karen (2019) (AGU OSM, Honolulu, HI) Feb.2022
- Evaluating Storm Surge Predictions for Hurricane Michael (AGU OSM, Honolulu, HI) Feb.2022
- Model Parameter Uncertainty in Tropical Storm Hydrodynamic Model Simulations (AGU OSM, Honolulu, HI) Feb.2022
- The Response of Mixed Layer Depth Due to Extreme Weather Events (AGU Fall Meeting, New Orleans, LA) Dec.2021
- Hydrodynamic and Thermohaline Effects on the Gulf due to Super Cyclone Gonu of 2007 (AGU OSM, San Diego, CA) Feb.2020
- Ensemble simulations of nearshore hydrodynamics and morphologic evolution (AGU-OSM, San Diego, CA) Feb.2020
- Ensemble simulations of hydrodynamic and morphologic change during extreme weather events (AGU Fall Meeting) Dec.2019
- Lee, W., and Kaihatu, J.M. (2018) Effects of Desalination on Hydrodynamic Process in Persian Gulf. Coastal Engineering Proceedings, 143(2), [S.I.], n.36, p. papers.3, dec. 2018. ISSN 2156-1028. Available at <https://journals.tdl.org/icce/index.php/icce/article/view/8396/7204>. doi: <https://doi.org/10.9753/icce.v36.papers.3>.

- Effect of Desalination on Hydrodynamic Process in Persian Gulf (ICCE 2018, Baltimore, MD) Aug.2018.
- Mixing and Effect of Desalination on Processes in Arabian Gulf (YCSECA 2016, Kingston, Canada) Jun.2016
- Modeling Salinity Changes in the Persian Gulf (AGU OSM, New Orleans, LA) Feb.2016

Scheduled

- Transferability and Forcing Sensitivity of a Multi-Domain SFINCS Framework for Hurricane-Driven Coastal Flooding across the U.S. Gulf Coast, (AGU Fall Meeting 2026, San Francisco, CA) Dec.2026
- Linking Multi-Event Pluvial Flood Hazards, Asset Impacts, and Adaptation in the Lower Rio Grande Valley, (AGU Fall Meeting 2026, San Francisco, CA) Dec.2026
- Annual Monitoring of Texas Colonias Using PlanetScope Time Series and Machine Learning, (AGU Fall Meeting 2026, San Francisco, CA) Dec.2026
- Beyond the Floodplain Boundary: Airborne LiDAR for Building-Level Flash-Flood Vulnerability in Kerrville, Texas, (AGU Fall Meeting 2026, San Francisco, CA) Dec.2026

Technical Reports and Others

- FIELD NOTES (2026) "Texas Applied Water Research Team", Bureau of Economic Geology's new biannual publication. <https://www.beg.utexas.edu/publications/field-notes>
- Penko., et al. (2020) "Direct Numerical Simulation of Fluid-Sediment Wave Bottom Boundary Layer", NRL FY2020 High Performance Computing System Report, U.S Naval Research Laboratory, NASA's John C. Stennis Space Center, MS.
- Lee., et al. (2021) "Model Validation: Hydrodynamics, Waves, and Storm Surge Predictions in Puerto Rico Using Delft3D Flexible Mesh", NRL Formal Report, U.S Naval Research Laboratory, NASA's John C. Stennis Space Center, MS.
- Veeramony., et al. (2021) "Storm Surge Modeling Using Delft3D FM - A Flexible Mesh Model", NRL Formal Report, U.S Naval Research Laboratory, NASA's John C. Stennis Space Center, MS.

Other Scholarly Contributions

- **Ph.D Dissertation:** W. Lee Jun.2018
 "Modeling of Three Dimensional Wind-Driven Hydrodynamic and Environmental Processes in the Arabian/Persian Gulf"
 Zachry Dept. of Civ. Engrg., TAMU
 Supervisor: Prof. James M. Kaihatu
 Committee Members: Profs. Scott A. Socolofsky, Vijay Panchang, Alejandro Orsi

A three-dimensional hydrodynamic model has been developed to investigate thermohaline processes, ocean circulation driven by winds, and other hydrodynamic features in the Arabian/Persian Gulf area. The validated model can be used to study a wide array of effects and processes in the region, with potential application to future climate scenarios and the resultant effects on oceanic processes. This dissertation centers around the development of a modeling system which can provide reliable, accurate and meaningful results of ocean hydrodynamics, waves and impacts of human use. The research goals were: 1. to develop and validate a three-dimensional hydrodynamic model for the Arabian/Persian Gulf, 2. to study various processes (the effects of desalination process and extreme condition) on the thermohaline characteristics, and 3. to discuss the future research with the effect of climate change, which can affect established weather patterns and, consequently, the ocean circulation and environment in the Gulf region.

- **M.E. Thesis:** W. Lee Dec.2012
 "The Interaction of the Moored Pontoon-Type Floating Breakwater with a Horizontal Plate"
 Dept. of Civ. Engrg., SNUST
 Supervisor: Prof. SungTae Kee

The performance of pontoon-type floating breakwater with a horizontal plate under the bottom of pontoon is identified through established/developed analytic theory and multi-domain boundary element method, and to figure out the optimal design from an inclusive parametric study. The validity of numerical analysis is confirmed in the energy relation and investigated the transmission & reflection coefficients. The case of fixed pontoon is compared with the graph published by Abul-Azm & Gesraha (2000). Afterward, the dynamic motion is investigated from the various variable that the width, draft & mooring line stiffness of pontoon are regarded as the major parameters.

RESEARCH PROJECTS FUNDED: External (PI or listed co-PI)

Current and Pending Projects

- COAST-NISAR: Compound Flooding Assessment from Space: Tracking Natural and Anthropogenic Drivers with NISAR in the US Gulf Coast - A.3 NASA ISRO Synthetic Aperture Radar Mission Data, Applications, Research, and Technology Team, NNH25ZDA001N-NISAR, Co-PI, \$975,000, selected Sept.2026-Sept.2029

- Predicting Submonthly Gulf Coast Sea Level Variability and its Effect on Flooding, by National Academies Sciences Engineering Medicine - Gulf Research Program: Gulf Sea Level Variation and Rise (GSLVR) Cycle II)
Principal Investigator, \$1,500,000, Awarded in Oct. 2025 Dec.2025-Dec.2030
- Mapping Lower Rio Grande Valley Colonias Using Remote Sensing and Machine Learning Data Analytics. (by Texas General Land Office)
Co-Investigator, \$921,956, Join in Sept. 2025 Jan.2025-Dec.2027
- BEG JSG FAC & Research Scientist, Yearly Support that is non-project related funding which requires no committed effort (by BEG FRSA)
Principal Investigator Sept.2022-Aug.2026
- Pending Support - Modeling Surface-Groundwater Interactions during Surge and Compound Flooding (by USACE-ERDC Civil Works)
Lead Principal Investigator, \$1,500,000 TDB three-year project

In Progress

- (Full-Proposal) Texas Offshore Legacy and Orphaned Wells: Environmental and Climate-Informed Risk Assessment and Integrated Mapping: by National Academies Sciences, Engineering, Medicine (NASEM) Gulf Research Program (GRP), Co-PI, May. 2026 (submitted)

Past Projects

- Texas Imperative - Water System and Drought (by Bureau of Economic Geology - University of Texas at Austin)
Co-Investigator, \$750,000 annually - Project STARR, Join in Sept. 2025 Apr.2024-Aug.2026
- MuSiKAL: Multiphysics Simulations and Knowledge discovery through AI/ML technologies (by Dept. of Energy)
Co-Investigator, \$3,000,000 Oct.2021-Aug.2025
- Assessing Climatic Impacts on Wetland Transformation and Resilience along the Texas Coast using Remotely Sensed Imagery (by UTMSI Gulf Coast Impact 2075)
Co-Investigator, \$99,517.87 Sept.2024-Aug.2025
- Probabilistic Surge and Inundation Prediction System (PSIPS), ONR, U.S. Department of Defense, N0001420WX00786 and N0001420WX01407.
Post-Doctoral Researcher Dec.2018-Dec.2021
- The Response of Thermohaline Structures and Sonic Layer Depth to forcing from Extreme Weather Events (by NRC Research Associationship Program through the Naval Research Laboratory)
Principle Investigator, \$81,548, \$84,398, \$88,335 (annual stipend + \$3,000 annual travel budget) Dec.2018-Dec.2021
- Forecasting bathymetric evolution in shallow coastal regions: Linking operational prediction system with remote sensing, ONR, U.S. Department of Defense
Post-Doctoral Researcher Dec.2018-Dec.2019
- Multi-Scale Modeling of Hydrodynamics in Arabian Gulf (by National Priority Research Program in Qatar National Research Fund)
Ph.D Graduate Research Assistant Jun.2014-Aug.2018
- A parameterized climate change projection model for hurricane flooding, wave action, economic damages and population dynamics (by Sea Grant Program)
Ph.D Graduate Research Assistant Sept.2013-Feb.2014

PROFESSIONAL EXPERIENCES

Collaborative Research Groups

- Developing AI/ML Surrogate Storm Surge and Flood Model - with NVIDIA Researcher Kamyar Azizzadenesheli Jan.2024-present
- Multiphysics Simulations and Knowledge Discovery through AI/ML Technologies (MuSiKAL): Teams UT Austin, LSU, Univ. of Notre Dame, Pacific Northwest National Laboratory May.2022-Aug.2025
- Probabilistic Surge and Inundation Prediction System (PSIPS): Teams U.S Naval Research Laboratory, Deltares, U.S Geological Survey Jan.2020-Dec.2021

Technical Advising-Students

- Coastal Hazards Lab (Ph.D and M.S. Students, under Dr.Jun-Whan Lee) University of Texas at Austin - Maseeh Department of Civil, Architectural and Environmental Engineering: Introduction of Reduced-Complexity Model (SFINCS): Modeling lecture. <https://sites.utexas.edu/junwhanlee/> Sept.2024-Present
 - Hyunje Yang (Ph.D Student): simulating of storm surge and tsunami propagation, model configurations.
 - Enayet Chowdhury (Ph.D Student): simulating storm surge and compound flooding.

- Armando Ulises Santos Cruz (Ph.D Student): developing flood model and its validations.
- Aaron Murai (currently, M.S Student) Texas A&M University: Setting the Hydrodynamic and Wave coupling and testing prediction of hurricane surge and flooding along the Texas coast May.2025-Present
- Philip M Souza (currently, Post-doc) University of Texas at Austin - Marine Science Institute: Salinity and temperature modeling to understand how available spawning habitat might change during the coming decades Apr.2024-Present
- Javad Teymouri (currently, Ph.D) University of Texas at Arlington: SFINCS modeling setup and building model structure, compiling, running the models Dec.2025-Feb.2026
- Mark K. Wang (former, Ph.D student) University of Texas at Austin – Jackson Geoscience Department: SFINCS modeling and development in Texas Apr.2025-May.2025
- William S. Bailey (currently, Ph.D student) University of Texas at Austin – Jackson Geoscience Department: Develop the Delft3D FM morphological modeling, sediment transport and microplastic transportation Apr.2025-May.2025
- Xuan Ma (former, Ph.D Student) Texas A&M University: Setting the Hydrodynamic and Wave coupling and testing tutorial of each mode in coupling Mar.2023-Dec.2024
- Arrumi Isnandiahening (former, MS Student) State College of Meteorology Climatology and Geophysics, in Indonesia: Guide and advise the setting hydrodynamic modeling and basic functional approaches Nov.2023
- Jin Young Kim (former Ph.D Student) Texas A&M University: Setting the Hydrodynamic and Wave coupling, generating forcing conditions and running in HPC 2017-2018
- Szu-Ting Lee (former M.E Student) Texas A&M University: Setting the Hydrodynamic modeling and sensitivity study in HPC 2017-2018
- Chi Lu (former M.S Student) Texas A&M University: Setting the Hydrodynamic and morphodynamic modeling and sensitivity study in HPC 2015-2016
- Woongsuk Pae (former M.S Student) Texas A&M University: Setting the Hydrodynamic and morphodynamic modeling and HPC environment in simulations 2015-2016

Research Experience

- 1) *Research Assistant Professor - PI* at University of Texas, Austin, TX. May.2022-present
- Develop and evaluate the coupled modeling systems using the advanced modeling suites for accurate predictions of extreme weather events (surge, flooding and inundation, etc.).
 - Develop and incorporate various sources of model input in the coupled hydrodynamic-wave-land-atmospheric model under the extreme weather events.
 - Develop and incorporate various sources of model input in the probabilistic storm surge and flood-inundation modeling, using new fast simulation tool (ensemble approach).
 - Develop a Digital Twin framework capable of combining, in near- and real time, a multitude of data, distributed and parallel computing resources, coupled with Artificial intelligence (AI) and Scientific Machine Learning (SciML) techniques, to greatly strengthen current coastal hazard prediction skills, improving flood preparedness and mitigation capabilities.

Research Projects (involved)

- a) *Predicting Submonthly Gulf Coast Sea Level Variability and its Effect on Flooding* (National Academies Sciences Engineering Medicine - Gulf Research Program: Gulf Sea Level Variation and Rise (GSLVR) Cycle II) Dec.2025-Dec.2030

- Develop a coastal flood modeling and prediction framework linking ocean circulation, tides, and atmospheric forcing to submonthly sea level variability.
- Incorporate the various sources of bathymetry, meteo input sources in the modeling system.
- Quantify the influence of sea level anomalies on coastal flooding and compound surge-rainfall interactions.
- Validate probabilistic predictions against observed tide-gauge and flood-event records across the Gulf Coast.
- Incorporate reanalysis, altimetry, and in-situ observations to train ML-based predictive models of short-term sea-level fluctuations.

- b) *Mapping Lower Rio Grande Valley Colonias Using Remote Sensing and Machine Learning Data Analytics.* (Texas General Land Office)

Joined in Sept.2025-Present

- Apply high-resolution satellite imagery and LiDAR data to identify and classify colonias and flood-prone settlements across the Lower Rio Grande Valley.

- Develop ML-based feature extraction pipelines for land-use, impervious surfaces, and infrastructure quality assessment.
- Develop process-based flood modeling pipelines for LRGV regions.
- Provide data-driven insights to support state and local agencies in infrastructure prioritization and risk mitigation planning.

c) *Texas Imperative - Water System and Drought* (Bureau of Economic Geology - University of Texas at Austin)

Joined in Sept.2025-Present

- Developed the WAM-based hydrologic routing framework, reconstructing flow contributions between control points and integrating channel-loss coefficients (FCL) to replicate observed hydrograph patterns across multi-year simulations.
- Established a machine-learning correction pipeline (FNO/Latent-Spectral models) to learn and reduce residual biases between simulated and observed discharges, enabling physics-informed flow prediction and improved runoff forecasting accuracy.

d) *Multiphysics Simulations and Knowledge discovery through AI/ML technologies* (MuSiKAL, by Dept. of Energy)

May.2022-Aug.2025

- Develop the coupled modeling system for land-ocean-atmosphere to predict the surge, flooding and inundation.
- Incorporate the various sources of bathymetry, meteo input sources in the modeling system.
- Validate the model predictions of surge and inundation with field observations.
- Develop the robust and reliable Digital Twin Framework.
- Provide an automated system that will produce probabilistic forecasts of surge and flooding levels at significant areas.

e) *Assessing Climatic Impacts on Wetland Transformation and Resilience along the Texas Coast using Remotely Sensed Imagery* (by University of Texas Marine Science Institute Gulf Coast Impact 2075) Sept.2024-Aug.2025

- Develop the GEE tool and pairing reflectance data with meteorological-ocean data to assess the coupling between climate-driven stress and plant response.
- Evaluate Climate-Driven Vegetation Dynamics and Wetland Resilience on the Texas Coast using Remotely Sensed Imagery and Machine Learning.

2) *NRC Postdoctoral Fellow* at U.S Naval Research Laboratory, NASA's John C. Stennis Space Center, MS.

Dec.2018-Dec.2021

- Develop and evaluate the ensemble systems using the advanced modeling suite, Delft3D-FM and Delft3D 4 for accurate predictions of extreme weather events (surge and inundation).
- Develop and incorporate a coupled hydrodynamic-wave model for the thermohaline responses from extreme weather events.
- Handle various US-Navy ocean meteorological and oceanographic data for a numerical modeling system.
- Handle the NRL-High Performance Computing (practical parallel processing) and DOD-High Performance Computing Modernization Program (HPCMP).
- Identify the Mixed Layer Depth (MLD) from the model results to anticipate the variations of Sonic Layer Depth (SLD) during extreme weather events.
- Estimate the variabilities of MLD and SLD against the characteristics of extreme weather events.
- Apply the ENsemble Kalman Filter (ENKF) with LHS skill to predict bathymetric changes.

Research Projects (involved)

a) *Probabilistic Surge and Inundation Prediction System* (PSIPS, by Office of Naval Research) Jan.2020-Dec.2021

- Develop and incorporate the hydrodynamic model using Delft3D-FM suite for accurate surge and inundation predictions.
- Generate the beautiful unstructured grids, flexible meshes, for the Puerto Rico.
- Incorporate the various sources of bathymetry in the modeling system.
- Validate the modeling system's predictions of surge and inundation with field observations.
- Provide an automated system that will produce probabilistic forecasts of surge and inundation levels at operationally significant areas.

b) *Ensemble Simulations for Probabilistic Surge and Inundation Prediction* (a part of PSIPS) Jan.2020-Dec.2021

- Generate the beautiful unstructured flexible mesh grid, for the Gulf of Mexico, during the Hurricane Katrina 2005.
- Develop and incorporate the initial model-setup of Delft3D-FM during the Hurricane Katrina and Rita 2005.

- Validate the modeling system's predictions of water level and wave properties with field observations.
- Evaluate probabilistic system along with ensemble predictions.
- Provide a comparison of model-performance between Delft3D 4 and Delft3D FM suites.

c) *The response of thermohaline structures and sonic layer depth to forcing from extreme weather events* (by NRC Research Associateship Program, at NRL) Dec.2018-Dec.2021

- Develop and incorporate a three dimensional (Z-coordinate) hydrodynamic model to carry out the thermohaline predictions from extreme weather events in Gulf of Mexico.
- Incorporate various US-Navy ocean meteorological and oceanographic data into the modeling system.
- Identify the Mixed Layer Depth (MLD) from the model results to deduce the Sonic Layer Depth to vertical thermohaline structures during extreme weather events.
- Evaluate the variabilities of MLD and SLD against the characteristics of extreme weather events.

d) *Forecasting bathymetric evolution in shallow coastal regions: Linking operational prediction systems with remote sensing* (by Department of Defense, NRL) Jan.2019-Dec.2019

- Grasp the Ensemble Kalman Filter (EnKF) method with data assimilation.
- Develop the Ensemble Kalman Filter (EnKF) method with Latin Hyper-cube Sampling (LHS) technique to predict bathymetric evolution with a coupled hydrodynamic-wave-morphology model.

3) *Research Assistant* at Zachry Department of Civil Engineering, Texas A&M University, TX. Jun.2014-Aug.2018

- Develop a numerical modeling system and study the hydrodynamic predictions (Mixing process, forcing).
- Estimate and evaluate the environmental impacts of the desalination process and potential effects (Saline effluence).
- Study the risk and extreme values of extreme weather conditions (Cyclones, Hurricanes).
- Develop computationally efficient Wave Response Function (WRF) to predict maximum significant wave height.
- Evaluate future climate change and estimate the future possible changes of hydrodynamic predictions.

Research Projects (involved)

a) *Multi-scale modeling of hydrodynamics in Arabian Gulf* (by Qatar National Research Fund) Oct.2014-May.2018

- Develop a numerical modeling system for waves and hydrodynamics for the Gulf.
- Configure the modeling system to provide the various applications.
- Use models and measurements to characterize turbulence properties and microstructure in the water column and their impact on larger scale hydrographic properties and associated forcing.

b) *A parameterized climate change projection model for hurricane flooding, wave action, economic damages and population dynamics* (by Sea Grant Program) Sep.2013-Feb.2014

- Parameterize the maximum wave heights in coastal regions due to hurricanes.
- Analyze data of temporal and spatial hurricane data over Corpus Christi TX, Gulf Port MS, Panama City FL in USA.
- Develop the 'Modified SPM (Shore Protection Manual)' method to evaluate the future extreme value statistics analysis of maximum significant wave heights by hurricanes using the simulation of hurricanes via coupled ADCIRC-SWAN & ADCIRC-STWAVE.

4) *Research Assistant* at Department of Civil Engineering, Seoul National University of Science & Technology, Seoul, South Korea. Mar.2011-Feb.2012

- Develop a numerical model for a floating breakwater with a horizontal plate.
- Analyze the historic weather data in South Korea.

Teaching Experience

Teaching Assistant at Department of Civil Engineering, Seoul National University of Science & Technology, Seoul, South Korea. Mar.2010-Feb.2011

- Applied Engineering Mathematics.
- Fluid Mechanics.

Laboratory Experience

Lab. Experiment Assistant at Department of Civil Engineering, Seoul National University of Science & Technology, Seoul, South Korea. Sep.2009-Dec.2009

- Conduct Hydraulics Experiments (Wave maker, pipe flow, open channel, etc.).
- Manage and organize the laboratory experiment.

Invited Seminars

- Coastal Resilience through Integrated Modeling of Compound Flooding Events (Bureau of Economic Geology, University of Texas, Austin TX.)
<https://www.beg.utexas.edu/presentations/coastal-resilience-through-integrated-modeling-of-compound-flooding-events> Mar.2025
- Coastal Resilience through Integrated Modeling of Compound Flooding Events (Environmental and Water Resources Engineering and Center for Water and the Environment Seminar Series, University of Texas, Austin TX.)
https://drive.google.com/file/d/1aSlcH9pIvZ8Jc2bgD2w0BCIwgAJQ4Y4V/view?usp=drive_link Mar.2025
- Predictions of Hydrodynamics, Waves and Environmental Processes under the Extreme Weather Conditions using a Coupled Hydrodynamic-Wave Modeling System (Bureau of Economic Geology, University of Texas, Austin TX.) - Virtual Seminar. Mar.2022
- Modeling of Three-Dimensional Wind-Driven Hydrodynamic and Environmental Processes (U.S. Naval Research Laboratory, NASA's John C. Stennis Space Center, MS) - expenses paid. Jul.2018
- Parameterization of maximum wave heights in coastal regions due to hurricanes (OCEN681, TAMU) Feb.2015

Professional Training and CertificationsTraining

- Introduction to FloodAdapt (FloodAdapt, a tool for flood-related adaptation planning.) Apr.2026
- A Delft3D Flexible Mesh training in River Modeling (hydraulic and river morphodynamics, real-time control) Apr.2024
- A Delft3D Flexible Mesh training in Environmental Modeling (sediment, pollution, tracer transport and water quality, ecological transport) Mar.2024
- SFINCS (Super-Fast INundation of CoastS) Modeling (Overland Flood Modeling due to Compound Events) Apr.2023
- Google Earth Engine Workshop, Austin, TX Jan.2023
- XBeach Modeling-Basic course in Morphodynamic Modelling during Storm Conditions - Virtual Jun-Jul.2022
- Introduction to Delft3D-FM and Delta Shell (NASA's John C. Stennis Space Center, MS) Feb.2020
- North America Deltares Software Days 2019, Delft3D 4 and Delft3D-FM - Hydrodynamics and Morphodynamics (Baton Rouge, LA) Mar.2019
- WAVEWATCH III - Wave Summer School (NCEP/NOAA & University System of Maryland Foundation) Jul.2016

Certifications

- Texas Executive Education Program - Bureau of Economic Geology & McCombs School of Business Leadership Program at UT Austin Oct.-Dec.2025
- UTLearn-Compliance & Diversity, Equity, and Inclusion Initiatives at UT Austin Nov.2024
- UTLearn-Compliance & Ethics Program at UT Austin May.2022
- UTLearn-Equal Employment Opportunity (EEO) May.2022
- Deltares' Course Certificate - Delft3D-FM and Delta Shell Feb.2020
- WAVEWATCH III Course Certificate - Wave Summer School Jul.2016
- Engineer of Civil Engineering in Republic of Korea Jun.2009
- Computer Intellectual Skill (Microsoft Certificate Professional 2nd Level) Feb.2008

Professional SkillsModeling system

- Artificial Intelligence and Machine Learning-based Surrogate Modeling
- Deep Learning - Physics-Aware Neural Networks with Neural Operators
- D-FLOW Flexible Mesh Interactor (unstructured grid)

- Delft3D FM suite D-FLOW FM, D-WAVES (SWAN)
- Delft3D FM suite D-Morphodynamics, D-Water Quality
- Delft3D 4 suite FLOW: Hydrodynamic model, WAVE (SWAN)
- SFINCS (Super-Fast Inundation of CoastS) Overland Flood Modeling due to Compound Events
- XBeach Morpho-Hydrodynamic modeling during extreme weather events
- Tropical Cyclone Wind Statistical Estimation (TCWiSE) Tool

OpenEarth

- Generate the depth using the global bathymetry dataset, and astronomical boundary conditions.
- Develop the cyclone modeling with spiderweb-wind field.
- Provide the convenient and various tools to set up the model.
- Provide the post-processing of model output (ex. using the Google Earth, Matlab-based post-processing tools).

Modified SPM (Shore Protection Manual) Analysis

- Provide expedient calculation of maximum significant wave height forced by hurricane winds.
- Use in ensemble generation for risk evaluation.
- Call for significant less computational time with achieving much of accuracy.

Others

- Programming Languages: MATLAB, PYTHON, FORTRAN, JAVAscript.
- Engineering Software: MS Office, LaTeX, QGIS, ArcGIS, CityEngine, Cesium Digital Twin, Draw.io, Google Earth Engine.
- High Performance Computing: UNIX OS, TACC (Docker)
- Operating System: Window, Linux, Mac
- Languages: English, Korean

Selected Coursework

Coastal and Ocean: Ocean Wave Mechanics, Non-Linear Wave Dynamics, Fluid Dynamics for Ocean and Environmental Engineering, Environmental Fluid Mechanics (auditor), Coastal Engineering, Coastal Sediment Processes, Physical Oceanography, Oceanographic Data Methods and Graphical Representation, Dynamics of Offshore Structures

Numerical Modeling: Computational Fluid Dynamics

Applied Mathematics: Applied Partial Differential Equations, Applied Mathematics